Self-Propelled Arm System

Project presentation

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Introduction and Outline

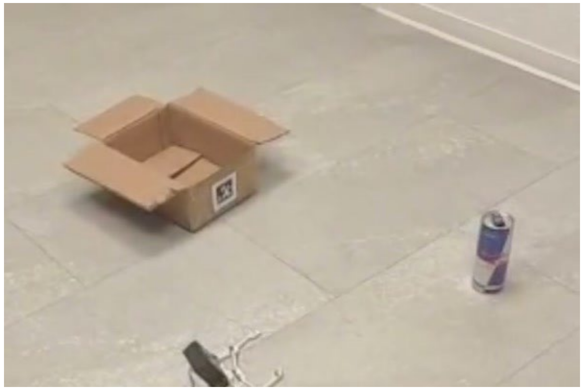
Overview

System summary

Autonomous can collection robot
Tracked indoor platform with a four link arm and a parallel gripper

A single RGB camera as the only sensor
No depth sensor, no range finder, no wheel encoders, no force sensing

Vision in the loop at every stage
Search, approach, grasp and delivery are decided from the image



Outline

Problem and platform
Task definition and hardware constraints

Architecture and control
System components, state machine, grasping

Perception and demonstration
Detection results and a complete run on hardware

Experimental results
Detector accuracy and mission timing

Limitations and conclusion
Open issues, summary and future work

Problem Statement

Objective and constraints

- Drink cans accumulate in indoor public spaces
- Collection is repetitive, low risk manual work
- The task combines perception, navigation, decision making and manipulation

Detection

Rotate in place until the detector accepts a target

Approach

Steer on the image error, stop on apparent size

Grasp

Lower the arm, push, close, lift

Delivery

Dock on the AprilTag and release above the bin

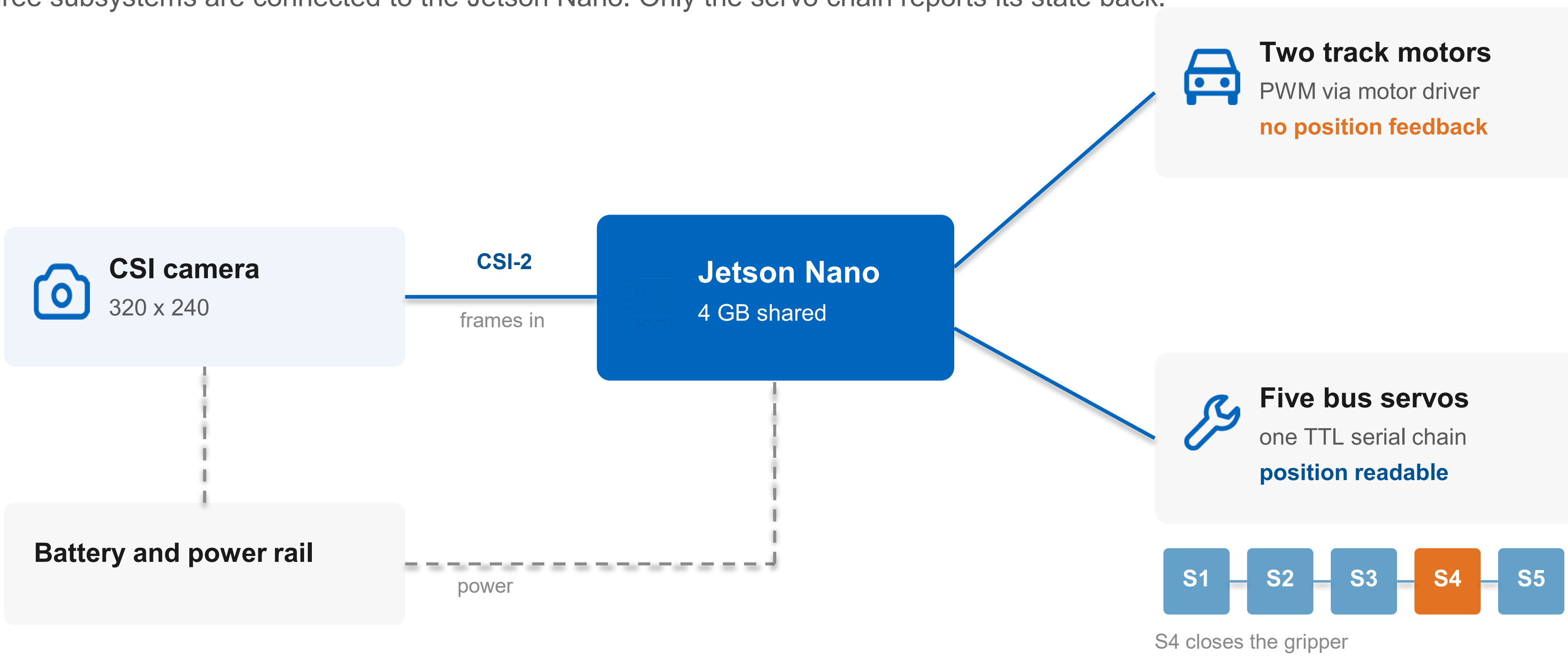
Sensor configuration

- **Single RGB camera**
- **No depth sensing**
- **No wheel encoders**
- **No force sensing**

System Architecture

Components and software framework

Three subsystems are connected to the Jetson Nano. Only the servo chain reports its state back.



Software stack

Mission layer
demo_core state machine, 16 states

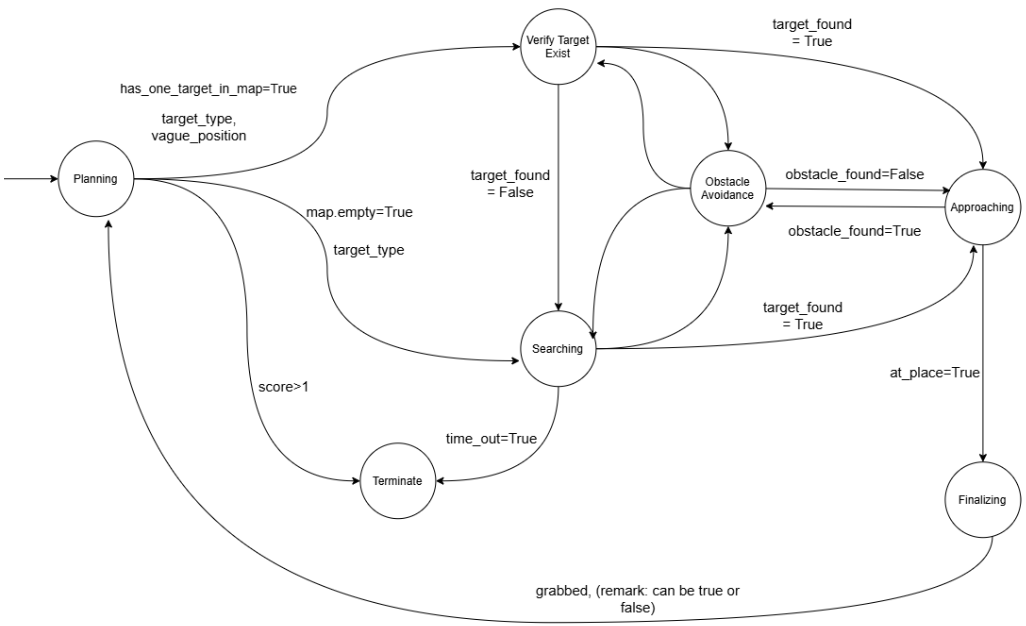
Perception layer
detectNet, pupil_apriltags, OpenCV

Hardware layer
jetbot Robot and Camera, TTLServo

Configuration
config.json, empirical_parameters.json

State Machine and Control Logic

Mission control



Design level state graph. Transition guards are fields of a single shared context object.

Vision driven states

Searching

Rotate until a target is accepted or the timeout expires

Aligning

Turn to tolerance, abort if the target is lost

Approaching

Steer or drive forward, stop on apparent size

Final verification

Tighter tolerance over consecutive frames

$$e_x = (x_c - W/2) / W$$

- Range -0.5 to 0.5, negative to the left of centre
- Stop thresholds: 0.36 box height, 0.15 tag height

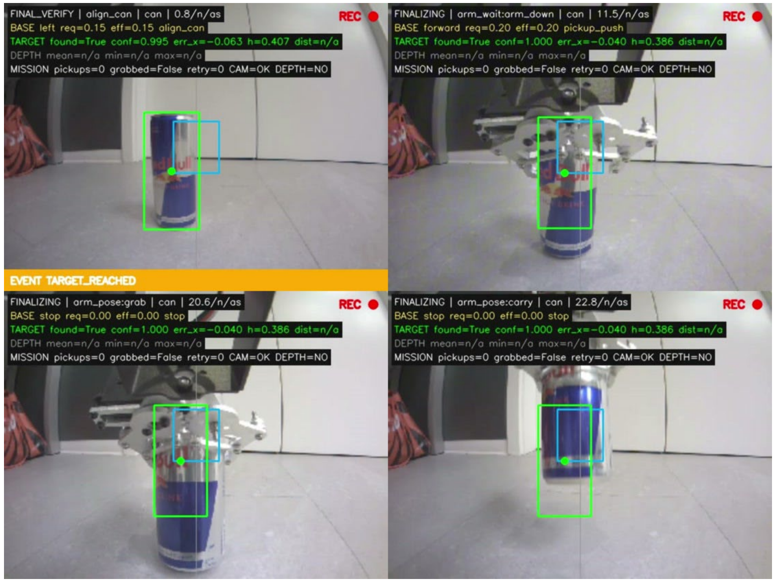
- Transitions are held in an explicit table, so an undeclared state and event pair raises an error
- Finalising requires a stable target, a condition map based navigation never reaches, so dead reckoning triggers no action

Grasping Strategy

Manipulation sequence

Fixed poses — no inverse kinematics

1	2	3	4	5	6
Wait	**Lower**	**Push**	**Hold**	**Close**	**Lift**
2.0 s settle	arm_down pose	Slow forward drive	5.0 s lock	Grab pose	Carry pose



Verify · push · close · carry

Why push, not reach

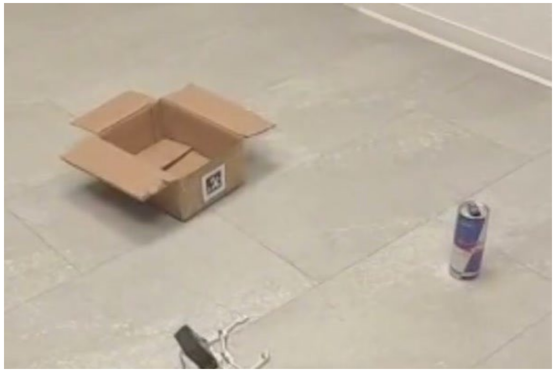
- Short arm, imprecise gripper
- Base drives reliably
- Can driven into open gripper
- Coarse motion beats fine manipulation

Servo feedback

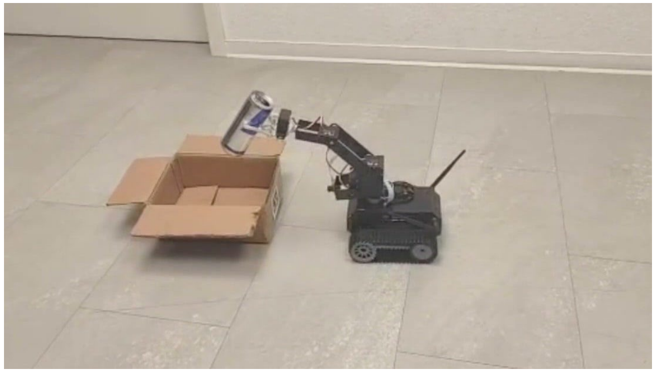
- Polled until readings stabilise
- Push blocked until settled
- 4.94 s settle · 7.0 s push · 5.0 s hold

Experimental Setup

Platform with grasped can



Arena: can, AprilTag bin



Release above bin

73.8 s
Full run

1
Camera only

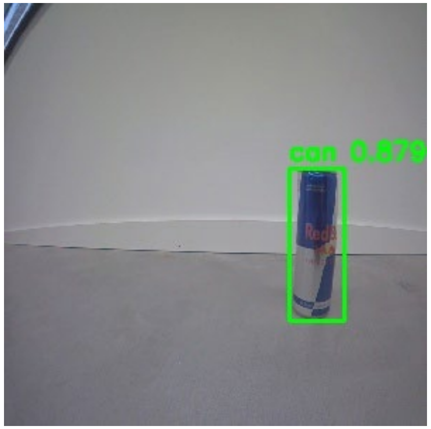
98
Can detections

41 / 59
Tag reads

Video

Object and Marker Detection

Two detectors, every controller input

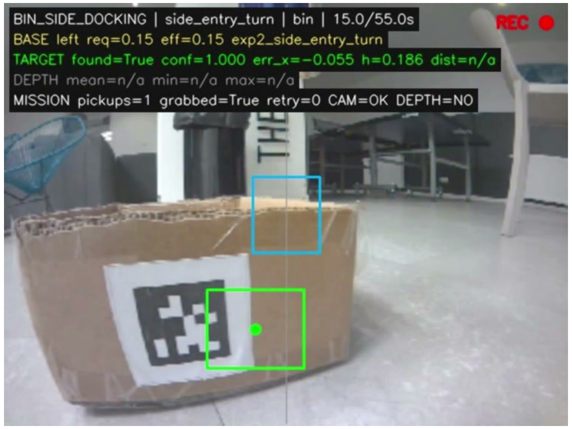


Detector output

SSD-MobileNet-V1 · 300 × 300

Can detection

error −0.119 · height 0.255

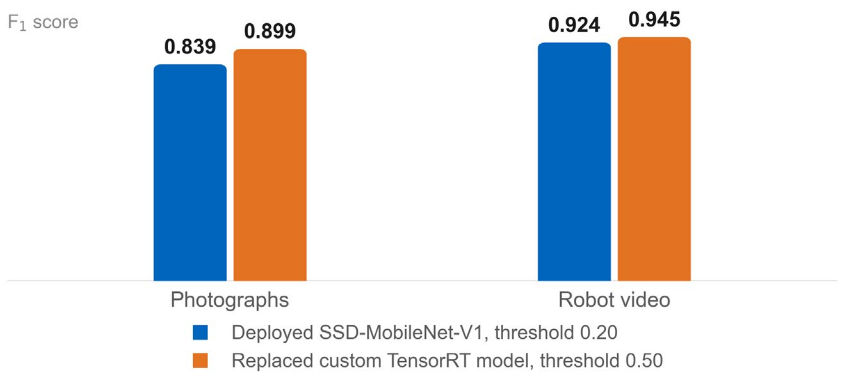


Tag detection

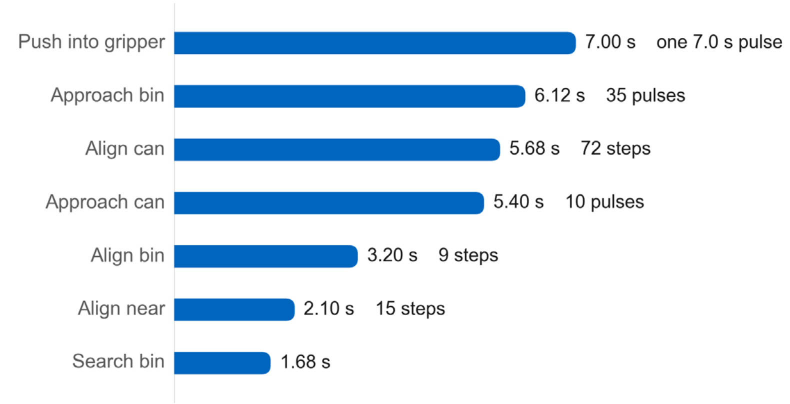
tag36h11 id 0 · height 0.186

Experimental Results

Detector accuracy, F1



Base motion per phase



73.8 s
Full mission

31.2 s
Base motion, 42 %

11.9 s
Deliberate holds

4.4×10^{-7}
PyTorch vs. ONNX

Accuracy vs. deployability

- −6 F1 photos, −2 video
- No custom post-processing
- Native NVIDIA path

Caveat: archived run

- Earlier prototype, TensorRT detector
- Legacy thresholds 0.400 / 0.500
- Counts and timings exact

Limitations and Challenges

Evaluation

Depth excluded

- 0.3 m general scenes
- 1 m white surfaces
- Detector reaches 3 m

Removed from control path

Pose drifts

- Left / right: 1–3 rad/s
- Modelled 0.471 / 0.942
- Error exceeds model

AprilTag: only anchor

Silent failure

- Verification disabled
- Pickup always recorded
- Empty gripper reports success

First fix: verification flag

Missing capability: relocalisation

Principal challenges

Open-loop turning unreliable
Pulses + visual correction

No gripper feedback
Full and empty identical

10–15 fps
Pulsed motion, low speeds

Model deployment
Accuracy traded for reliability

Depth underperformed
Unreliable at useful ranges

Grasp timing
Polling beat fixed delay

Conclusion and Future Work

Summary

Summary

- Mobile manipulator, single camera
- Closed on appearance, not geometry
- Errors degrade, never break
- Drifting map lengthens search
- Asymmetric turning needs corrections

Next steps

- **Enable grasp verification**
- **Calibrate turn constants**

Sensors worth adding

Forward distance sensor
First metric measurement

Gripper force sensing
Fixes empty gripper

Wheel odometry
Map becomes measurement

More room tags
Cheap relocalisation anchors